THE TWO FIBRE CYCLE CLASSES OF A COHERENT SHEAF ON A PRODUCT WITH THE PROJECTIVE LINE ARE RATIONALLY EQUIVALENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Local finiteness of the components of the support of a coherent sheaf. Same chapter chow.tex, label lemma-length-finite, invoked at proof line 3499. Makes the family of irreducible components W_i of the support of F of dimension k + 1 locally finite, so that the cycle [F]_{k+1} is a well defined locally finite sum.
- Rational equivalence of a family of fibres. Same chapter chow.tex, label lemma-rational-equivalence-family, invoked at proof line 3505. Converts the equality [F_0]_k = sum n_i[(W_i)_0]_k and its analogue for the fibre at infinity into the asserted rational equivalence between the two fibre classes.
- Additivity of divisors for a restricted family. Same chapter chow.tex, label lemma-additivity-divisors-restricted, invoked at proof line 3536. Supplies the final identity comparing length_A(M/tM) with the sum over the minimal primes of Ass(M) of the residue length times the local length, which is where the coefficient comparison ends.

Source reference. Original lemma 02QG: sources/stacks-chow/pinned-original-source.tex, source lines 373–373. Navigation pointer for \ref{lemma-additivity-divisors-restricted} used inside the counted span (line 3537); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QW: sources/stacks-chow/pinned-original-source.tex, source lines 1702–1702. Navigation pointer for \ref{lemma-length-finite} used inside the counted span (line 3499); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02S4: sources/stacks-chow/pinned-original-source.tex, source lines 3335–3335. Navigation pointer for \ref{lemma-rational-equivalence-family} used inside the counted span (line 3505); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 3471); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be a scheme locally of finite type over S. Let $\mathcal{F}$ be a coherent sheaf on $X \times \mathbf{P}^1$. Let $i_0, i_\infty : X \to X \times \mathbf{P}^1$ be the closed immersion such that $i_t(x) = (x, t)$. Denote $\mathcal{F}_0 = i_0^*\mathcal{F}$ and $\mathcal{F}_\infty = i_\infty^*\mathcal{F}$. Assume*

1. *$\dim_\delta(Supp(\mathcal{F})) \leq k + 1$,*
2. *$\dim_\delta(Supp(\mathcal{F}_0)) \leq k$, $\dim_\delta(Supp(\mathcal{F}_\infty)) \leq k$, and*
3. *for any embedded associated point ξ of $\mathcal{F}$ either $\xi \notin (X \times \mathbf{P}^1)_0 \cup (X \times \mathbf{P}^1)_\infty$ or $\delta(\xi) < k$.*

Then $[\mathcal{F}_0]_k \sim_{rat} [\mathcal{F}_\infty]_k$ as k-cycles on X.

Proof. Let $\{W_i\}_{i \in I}$ be the collection of irreducible components of $\mathrm{Supp}(\mathcal{F})$ which have δ-dimension $k + 1$. Write

$$[\mathcal{F}]_{k+1} = \sum n_i [W_i]$$

with $n_i > 0$ as per definition. Note that $\{W_i\}$ is a locally finite collection of closed subsets of $X \times_S \mathbf{P}^1_S$ by Lemma 02QW. We claim that

$$[\mathcal{F}_0]_k = \sum n_i [(W_i)_0]_k$$

and similarly for $[\mathcal{F}_\infty]_k$. If we prove this then the lemma follows from Lemma 02S4.

Let $Z' \subset X$ be an integral closed subscheme of δ-dimension k. To prove the equality above it suffices to show that the coefficient n of $[Z']$ in $[\mathcal{F}_0]_k$ is the same as the coefficient m of $[Z']$ in $\sum n_i[(W_i)_0]_k$. Let $\xi' \in Z'$ be the generic point. Set $\xi = (\xi', 0) \in X \times_S \mathbf{P}^1_S$. Consider the local ring $A = \mathcal{O}_{X \times_S \mathbf{P}^1_S, \xi}$. Let $M = \mathcal{F}_\xi$ as an A-module. Let $t \in A$ be the element cutting out $X \times_S D_0$ (i.e., the coordinate of $\mathbf{P}^1$ at zero pulled back). By our choice of $\xi' \in Z'$ we have $\delta(\xi) = k$ and hence $\dim(\mathrm{Supp}(M)) = 1$. Since ξ is not an associated point of $\mathcal{F}$ by assumption (3) we see that M is a Cohen-Macaulay module. Since $\dim_\delta(\mathrm{Supp}(\mathcal{F}_0)) = k$ we see that $\dim(\mathrm{Supp}(M/tM)) = 0$ which implies that t is a nonzerodivisor on M. Finally, the irreducible closed subschemes W_i passing through ξ correspond to the minimal primes $\mathfrak{q}_i$ of $\mathrm{Ass}(M)$. The multiplicities n_i correspond to the lengths $\mathrm{length}_{A_{\mathfrak{q}_i}} M_{\mathfrak{q}_i}$. Hence we see that

$$n = \mathrm{length}_A(M/tM)$$

and

$$m = \sum \mathrm{length}_A(A/(t, \mathfrak{q}_i)A)\mathrm{length}_{A_{\mathfrak{q}_i}} M_{\mathfrak{q}_i}$$

Thus the result follows from Lemma 02QG. □